universität
innsbruck
Department of
Computer Science

Recommender Systems

Organization

Eva Zangerle

CUSTOMERS WHO BOUGHT THIS ITEM:
ALSO BOUGHT:
piccolo
demotivateur.fr

Goals and Learning Outcomes

- Understand the basic principles of recommender systems, but also state-of-the-art recommender systems
- Understand and discuss current recommender systems publications and literature
- Implement recommender systems from scratch, but also using existing libraries
- Ability to evaluate, tune and analyze recommender systems and their performance
- Ability to carry out reproducible experiments
- Understand the strength, limitations, and weaknesses of recommender system approaches

Syllabus: Main Blocks

1️⃣ Foundations

- Problem formulation & feedback modeling
- Popularity, long tail, sparsity
- Implicit feedback & baselines

2️⃣ Personalization & Ranking

- Content-based
- Collaborative filtering and matrix factorization
- Learning to Rank
- Offline evaluation

Syllabus: Main Blocks cont.

3️⃣ Bias & Causal Evaluation

- Exposure & popularity bias
- Feedback loops
- Counterfactual evaluation

4️⃣ Modern Architectures

- Neural & two-tower models
- Sequential / transformer recommenders

Syllabus: Main Blocks cont.

5️⃣ Beyond Accuracy

- Diversity & novelty
- Multi-objective optimization
- Explanations & user control

Course Material

- Slides (OLAT)
- Current blogs and papers on relevant topics (slides/OLAT)

Course Material

- Practical Recommender Systems: Kim Falk; Manning Publications, 2019.
- Recommender Systems Handbook; Francesco Ricci, Lior Rokach, Bracha Shapira; Springer, 3rd edition 2022.
- Recommender Systems: The Textbook; Charu C. Aggarwal; Springer, 2016.
- Recommender Systems – An Introduction; Dietmar Jannach, Markus Zanker, Alexander Felfernig, Gerhard Friedrich; Cambridge University Press; 2012.
- Further material during reading sessions and on slides

Organization

- We will split our time in three parts:
  - Lecture
  - Reading and discussion
  - Tutorial
- VU3 is awarded 5 ECTS: approximately 6 hours at home, besides presence in class.

Coding

- You will have to code in Python for the tutorial part.
- Mostly Jupyter notebooks.
- We will make use of a variety of libraries for recommender systems, but also data analysis.

Grading

- Immanent character (you might only miss 2 weeks)
- Final grade based on
  - Written exam on lecture parts
  - Participation in reading discussions during class
  - Solutions and presentations of tutorial exercises
- See guidelines in OLAT

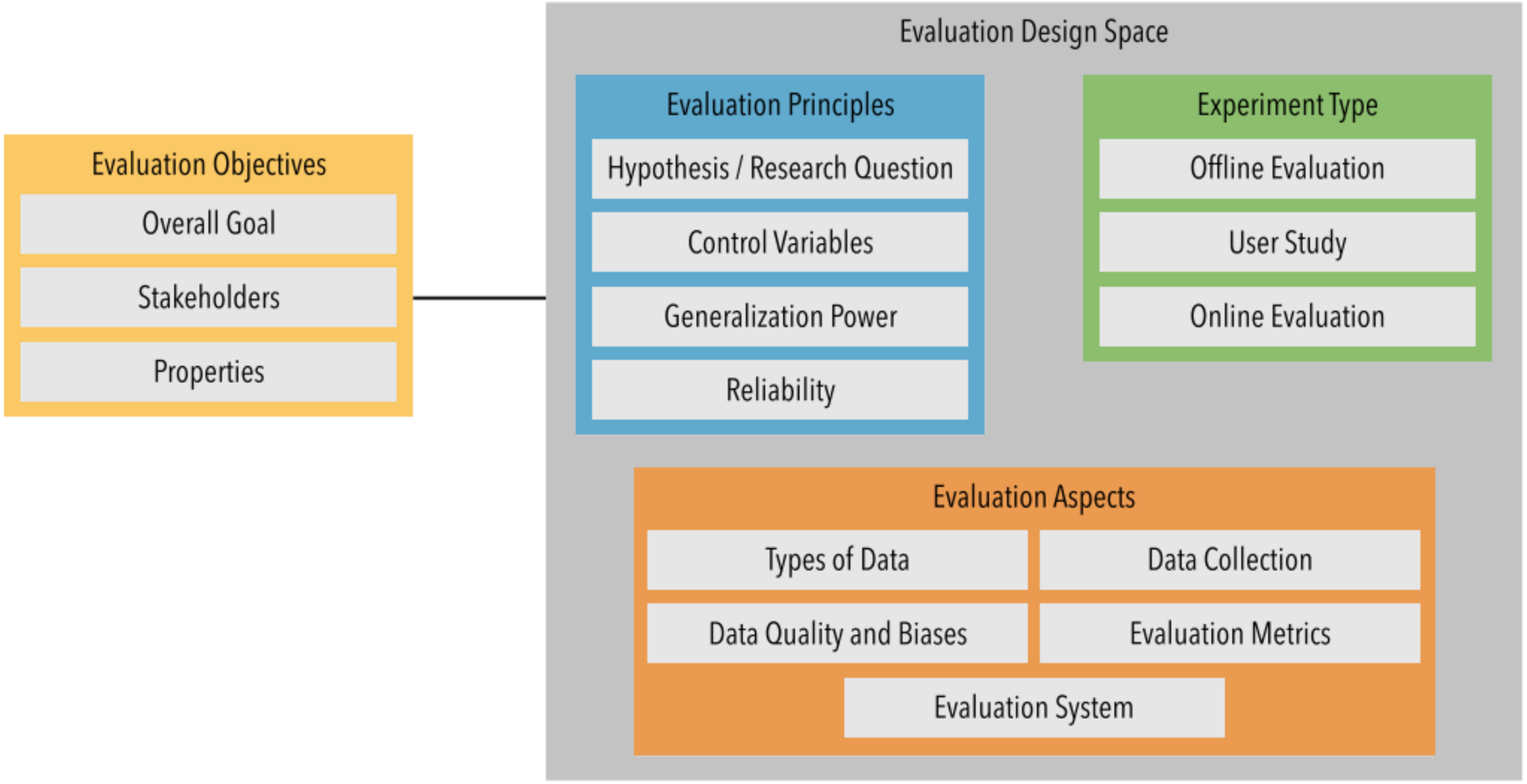
Questions, Feedback, ...

- Before, during, after course
- Email: eva.zangerle@uibk.ac.at

My research

- Recommender Systems
- User Modeling
- Music Information Retrieval
- Sometimes wandering off to Natural Language Processing, Databases and Indexing

Framework for Evaluating RecSys (FEVR)



Eva Zangerle and Christine Bauer. 2022. Evaluating Recommender Systems: Survey and Framework. ACM Comput. Surv. 55, 8, Article 170 (August 2023), 38 pages.

SPARE: Shortest Path Global Item Relations for Efficient Session-based Recommendation

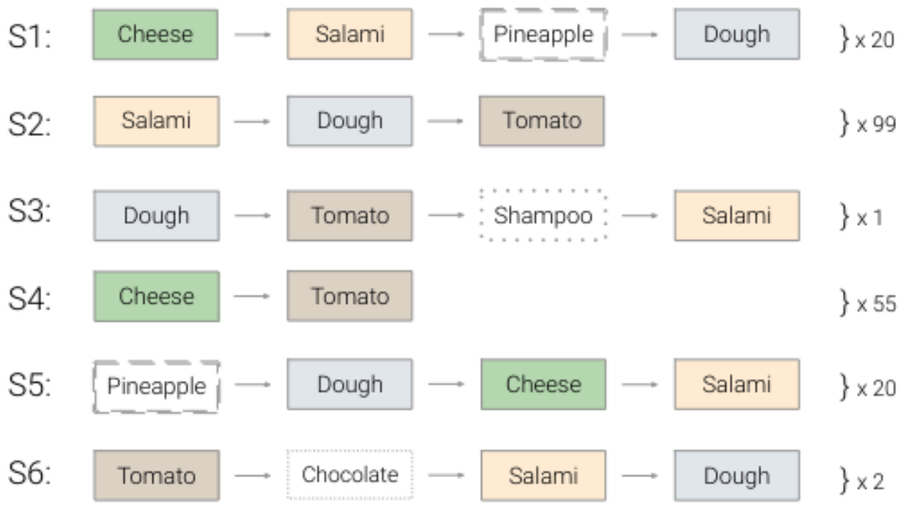


Figure 1: A toy example of an e-commerce grocery retailer scenario. Numbers indicate the frequency of each session.

Andreas Peintner, Amir Reza Mohammadi, and Eva Zangerle. 2023. SPARE: Shortest Path Global Item Relations for Efficient Session-based Recommendation. In Proceedings of the 17th ACM Conference on Recommender Systems (RecSys '23). Association for Computing Machinery, New York, NY, USA, 58–69. https://doi.org/10.1145/3604915.3608768

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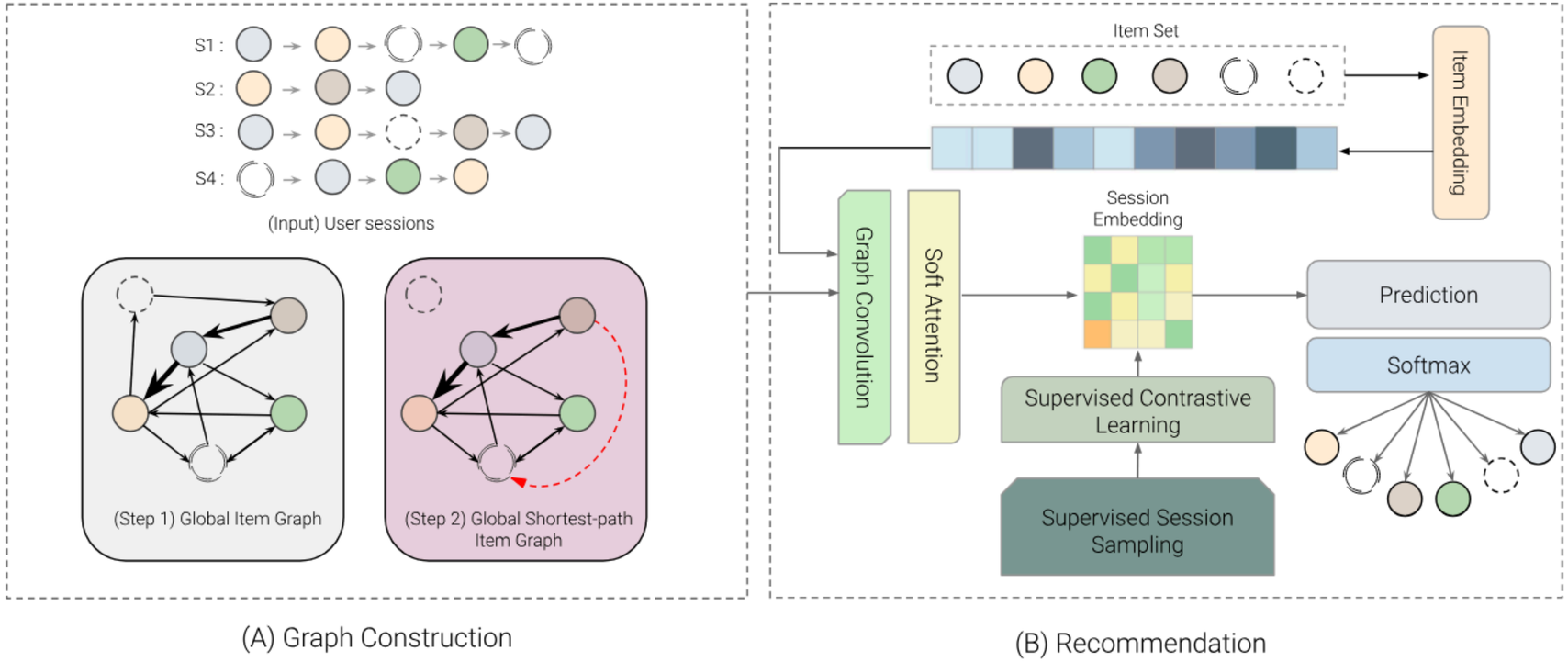

Figure 2: An overview of the global item graph construction and the pipeline of the proposed SPARE model.

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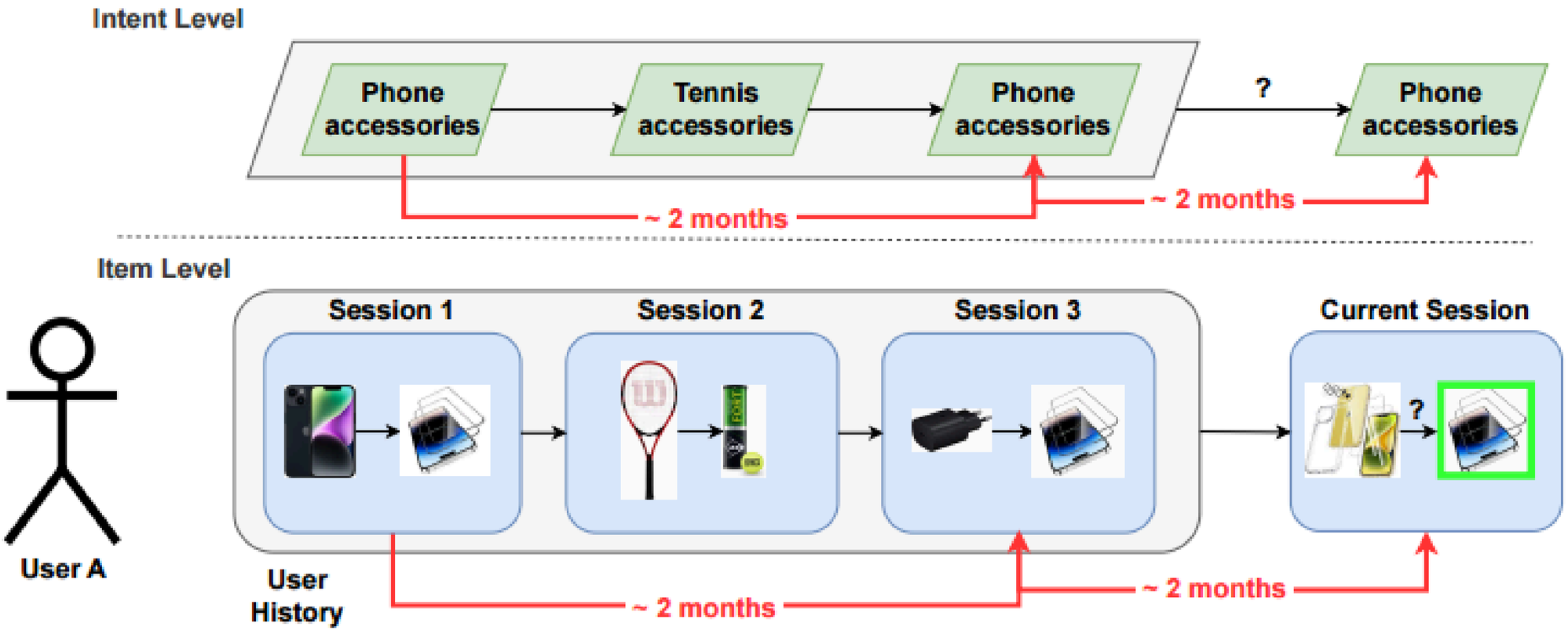
Hypergraph-based Temporal Modelling of Repeated Intent for Sequential Recommendation



Andreas Peintner, Amir Reza Mohammadi, Michael Müller, and Eva Zangerle. 2025. Hypergraph-based Temporal Modelling of Repeated Intent for Sequential Recommendation. In Proceedings of the ACM on Web Conference 2025 (WWW '25). Association for Computing Machinery, New York, NY, USA, 3809–3818.

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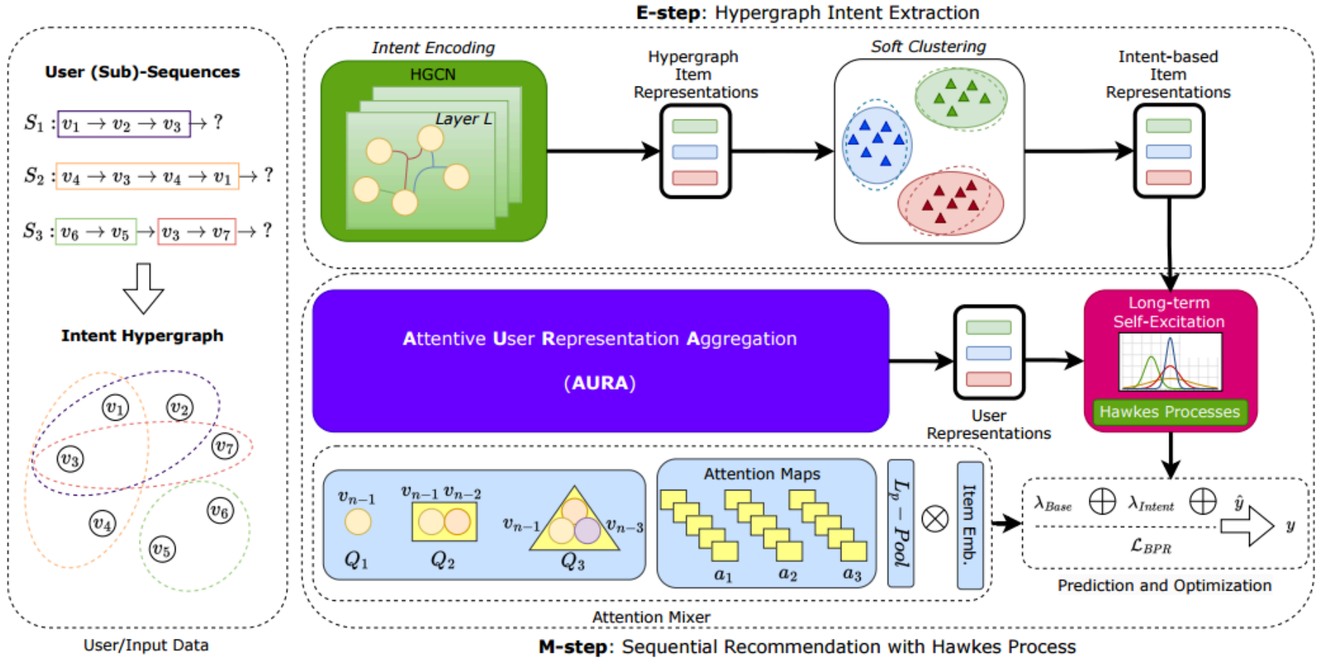
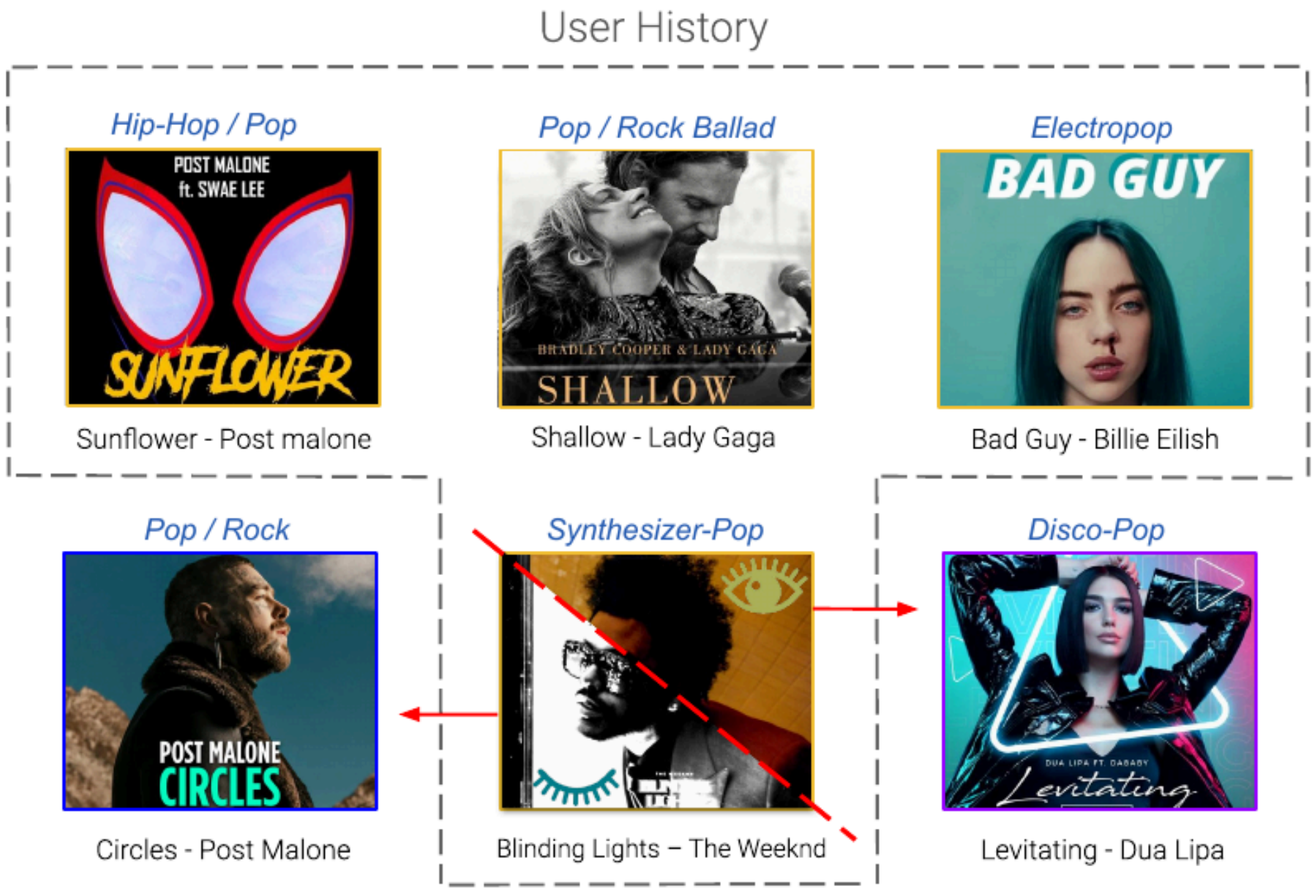


Figure 3: Overall architecture of HyperHawkes: In the E-step of the EM algorithm, our approach extracts latent intent representations via soft clustering of hypergraph-based item embeddings. In the M-step, we compute long-term user preference scores via Hawkes Processes based on the user base excitation from an attentive FISM and self-exciting effects of intents. We fuse short-term scores from the attention-mixture network and the long-term scores to get the preference score of the user for an item.

Andreas Peintner, Amir Reza Mohammadi, Michael Müller, and Eva Zangerle. 2025. Hypergraph-based Temporal Modelling of Repeated Intent for Sequential Recommendation. In Proceedings of the ACM on Web Conference 2025 (WWW '25). Association for Computing Machinery, New York, NY, USA, 3809–3818.

Beyond Top-1: Addressing Inconsistencies in Evaluating Counterfactual Explanations for Recommender Systems



Amir Reza Mohammadi, Andreas Peintner, Michael Müller, and Eva Zangerle. 2025. Beyond Top-1: Addressing Inconsistencies in Evaluating Counterfactual Explanations for Recommender Systems. In Proceedings of the Nineteenth ACM Conference on Recommender Systems (RecSys ’25). Association for Computing Machinery, New York, NY, USA, 515–520.

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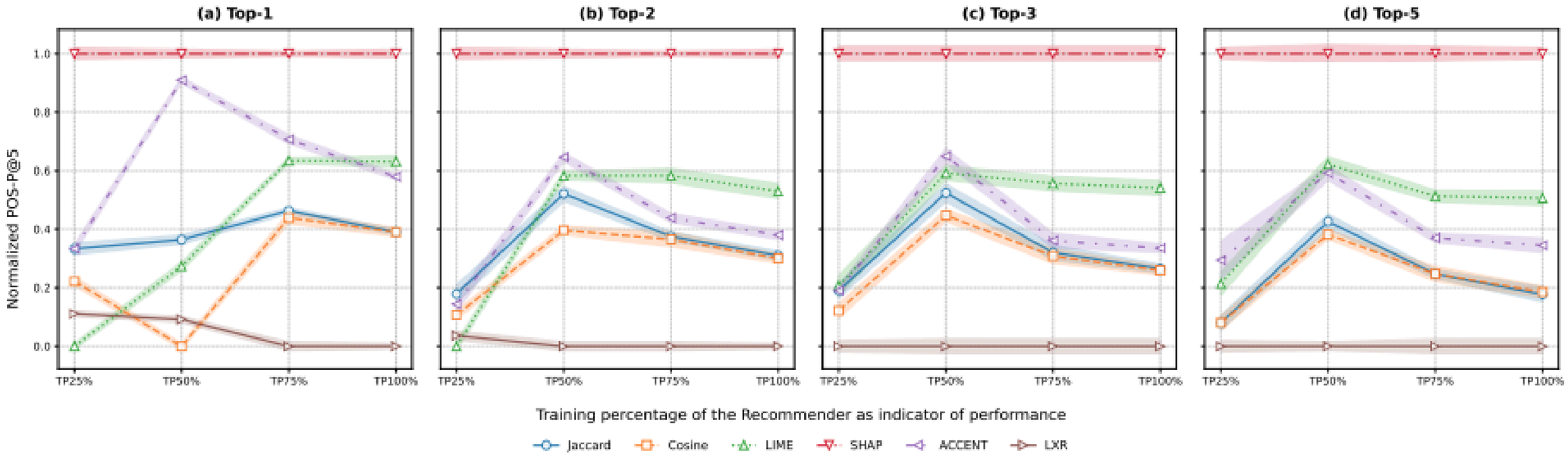

Figure 1: Comparison of CE methods based on POS@5 ↓ (lower value is the better) across four performance levels of the VAE recommender on the ML-1M dataset. The figure shows the impact of going beyond top-1 (a) and considering top-k (b-d) recommendations on improving consistency when evaluating CE models. To facilitate clearer comparisons, the values are normalized using Min-max normalization, and shading is used to represent the variance in the results.

Amir Reza Mohammadi, Andreas Peintner, Michael Müller, and Eva Zangerle. 2025. Beyond Top-1: Addressing Inconsistencies in Evaluating Counterfactual Explanations for Recommender Systems. In Proceedings of the Nineteenth ACM Conference on Recommender Systems (RecSys ’25). Association for Computing Machinery, New York, NY, USA, 515–520.